

Reproducibility supplement

Do MOS Fusion Weights Select Useful Embeddings? A System-Disjoint Attribution Audit

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This document accompanies the machine-readable CSVs and executable analysis modules. It reports all 40 attribution cells, all 16 discordant cells with uncertainty and refit frequency, all 120 Rashomon coordinate intervals, system-disjoint selection by corpus and budget, and the null controls. Values are copied from the canonical committed CSVs; `scripts/build_public_tables.py` produces text equivalents.

Data and pretrained checkpoints are not redistributed. See `metadata/dataset_splits.md` and `metadata/checkpoints.md` for sources, versions, licenses, and preprocessing instructions. The public artifact begins from already extracted representations.

Integrity command: `python scripts/verify_release.py`. It checks SHA-256 hashes, schemas, cross-file keys, four corpora, 40 attribution cells, all 1,023 masks in every released lattice, 50 training refits per cell, Rashomon solver success, nested folds, null families, table generation, and a synthetic end-to-end smoke test.

Table S1 — All attribution cells (1/2)

Gate is the full-coalition allocation. Singleton, LOO, and Shapley are held-out MSE utilities; positive means lower MSE. Train +freq is the fraction of training-system refits with positive Shapley. Test CI is a system bootstrap with fitted models fixed.

Corpus	Expert	Gate	Singleton	LOO	Shapley	CI low	CI high	Train +freq
brspeech	auditory_erb	0	0.4721	-1.01e-09	0.0514	0.0143	0.0908	0.6400
brspeech	beats	6.36e-18	0.3614	-7.28e-10	0.0488	-0.0111	0.1088	0.5000
brspeech	ced	0	0.5884	3.28e-08	0.1243	0.0644	0.1839	0.9800
brspeech	contentvec12	0.1532	0.2955	0.0238	0.0885	0.0030	0.1801	0.8800
brspeech	egemaps	0.5149	0.6512	0.0152	0.1693	0.0585	0.2870	1.0000
brspeech	rmvpe_cont	1.39e-17	-0.0065	9.01e-09	0.0042	-0.0352	0.0416	0.9800
brspeech	rmvpe_quant	2.81e-17	-0.0776	6.97e-08	-0.0033	-0.0191	0.0237	0.3800
brspeech	speaker	1.80e-16	0.5518	-9.53e-08	0.1202	0.0826	0.1753	1.0000
brspeech	wavlm	0.3021	0.4876	0.0238	0.1295	0.0485	0.2125	0.9200
brspeech	whisper	0.0298	-0.3430	-0.0048	-0.0486	-0.1261	0.0476	0.3800
bvcc	auditory_erb	3.70e-17	0.1782	-9.36e-08	0.0232	0.0118	0.0374	1.0000
bvcc	beats	0.2003	0.5651	0.0106	0.1137	0.0896	0.1402	1.0000
bvcc	ced	0.2222	0.5682	0.0139	0.1158	0.0909	0.1415	1.0000
bvcc	contentvec12	0.0614	0.4625	0.0023	0.0794	0.0576	0.1027	1.0000
bvcc	egemaps	0	0.3386	-9.87e-09	0.0500	0.0334	0.0684	1.0000
bvcc	rmvpe_cont	2.35e-17	0.0503	-1.04e-07	0.0068	0.0010	0.0131	1.0000
bvcc	rmvpe_quant	0	0.0211	-1.01e-07	0.0034	-0.0019	0.0091	0.5200
bvcc	speaker	0.0994	0.5004	0.0014	0.0858	0.0629	0.1104	1.0000
bvcc	wavlm	0.1672	0.5347	0.0016	0.0939	0.0701	0.1190	1.0000
bvcc	whisper	0.2495	0.4067	-0.0024	0.0656	0.0430	0.0894	1.0000

Table S1 — All attribution cells (2/2)

Gate is the full-coalition allocation. Singleton, LOO, and Shapley are held-out MSE utilities; positive means lower MSE. Train +freq is the fraction of training-system refits with positive Shapley. Test CI is a system bootstrap with fitted models fixed.

Corpus	Expert	Gate	Singleton	LOO	Shapley	CI low	CI high	Train +freq
singmos	auditory_erb	1.01e-17	0.0418	-1.54e-08	0.0181	0.0066	0.0290	1.0000
singmos	beats	0.1525	0.0983	0.0014	0.0375	0.0159	0.0598	0.9800
singmos	ced	0.2314	0.1725	0.0109	0.0605	0.0347	0.0859	1.0000
singmos	contentvec12	0.0761	-0.0919	-0.0025	-0.0005	-0.0214	0.0195	0
singmos	egemaps	0.2129	0.1738	0.0117	0.0579	0.0418	0.0736	1.0000
singmos	rmvpe_cont	7.00e-18	0.0092	2.87e-08	0.0111	0.0047	0.0177	1.0000
singmos	rmvpe_quant	4.21e-18	-0.0681	5.89e-08	0.0027	-0.0109	0.0143	0.2400
singmos	speaker	0.0078	0.1052	0.0002	0.0324	0.0232	0.0411	1.0000
singmos	wavlm	0.2156	0.0782	0.0037	0.0321	0.0142	0.0495	1.0000
singmos	whisper	0.1036	-0.2350	0.0030	-0.0113	-0.0520	0.0255	0
tmhintqi	auditory_erb	0	0.1861	3.51e-09	0.0222	0.0086	0.0388	1.0000
tmhintqi	beats	0.2710	0.5989	0.0058	0.1035	0.0543	0.1709	1.0000
tmhintqi	ced	0.2837	0.5920	0.0002	0.0964	0.0466	0.1656	1.0000
tmhintqi	contentvec12	0.0716	0.5263	0.0003	0.0805	0.0327	0.1477	1.0000
tmhintqi	egemaps	0	0.4858	2.20e-08	0.0695	0.0255	0.1323	1.0000
tmhintqi	rmvpe_cont	3.08e-17	0.0293	4.94e-09	0.0023	0.0008	0.0040	0.8800
tmhintqi	rmvpe_quant	7.52e-19	0.0292	2.74e-09	0.0032	-0.0027	0.0093	0.1800
tmhintqi	speaker	0.0118	0.4944	-1.19e-06	0.0724	0.0334	0.1266	1.0000
tmhintqi	wavlm	0.1697	0.6062	0.0065	0.1064	0.0580	0.1736	1.0000
tmhintqi	whisper	0.1922	0.5903	0.0091	0.1040	0.0605	0.1646	1.0000

Table S2 — Zero-gate / positive-Shapley cells (1/1)

The full set of 16 discordant cells. Gate zero uses the predeclared threshold <1e-6. Frozen zero freq is the frequency of a zero gate across frozen-expert system bootstraps; train +freq is the frequency of positive Shapley across refits.

Corpus	Expert	Gate	LOO	Shapley	CI low	CI high	Frozen 0 freq	Train +freq
brspeech	auditory_erb	0	-1.01e-09	0.0514	0.0143	0.0908	0.9850	0.6400
brspeech	beats	6.36e-18	-7.28e-10	0.0488	-0.0111	0.1088	0.9200	0.5000
brspeech	ced	0	3.28e-08	0.1243	0.0644	0.1839	0.6050	0.9800
brspeech	rmvpe_cont	1.39e-17	9.01e-09	0.0042	-0.0352	0.0416	0.8900	0.9800
brspeech	speaker	1.80e-16	-9.53e-08	0.1202	0.0826	0.1753	0.5900	1.0000
bvcc	auditory_erb	3.70e-17	-9.36e-08	0.0232	0.0118	0.0374	1.0000	1.0000
bvcc	egemaps	0	-9.87e-09	0.0500	0.0334	0.0684	0.9900	1.0000
bvcc	rmvpe_cont	2.35e-17	-1.04e-07	0.0068	0.0010	0.0131	1.0000	1.0000
bvcc	rmvpe_quant	0	-1.01e-07	0.0034	-0.0019	0.0091	1.0000	0.5200
singmos	auditory_erb	1.01e-17	-1.54e-08	0.0181	0.0066	0.0290	0.9100	1.0000
singmos	rmvpe_cont	7.00e-18	2.87e-08	0.0111	0.0047	0.0177	1.0000	1.0000
singmos	rmvpe_quant	4.21e-18	5.89e-08	0.0027	-0.0109	0.0143	0.7450	0.2400
tmhintqi	auditory_erb	0	3.51e-09	0.0222	0.0086	0.0388	1.0000	1.0000
tmhintqi	egemaps	0	2.20e-08	0.0695	0.0255	0.1323	0.8200	1.0000
tmhintqi	rmvpe_cont	3.08e-17	4.94e-09	0.0023	0.0008	0.0040	1.0000	0.8800
tmhintqi	rmvpe_quant	7.52e-19	2.74e-09	0.0032	-0.0027	0.0093	0.9900	0.1800

Table S3 — Complete Rashomon intervals (1/4)

Coordinatewise minimum and maximum gate weights among mixtures whose OOF MSE is within the stated relative tolerance of the optimum. All solver-success flags are checked by `verify_release.py`; KKT diagnostics are in `results/rashomon_kkt_audit.csv`.

Corpus	Expert	Tol.	Opt.	Min	Max	Width	Min OK	Max OK
brspeech	auditory_erb	0.0010	0	0	0.0009	0.0009	yes	yes
brspeech	beats	0.0010	0	0	0.0019	0.0019	yes	yes
brspeech	ced	0.0010	1.67e-16	1.67e-16	0.0157	0.0157	yes	yes
brspeech	contentvec12	0.0010	0.1532	0.1225	0.1839	0.0614	yes	yes
brspeech	egemaps	0.0010	0.5149	0.4790	0.5503	0.0713	yes	yes
brspeech	rmvpe_cont	0.0010	1.08e-16	1.08e-16	0.0069	0.0069	yes	yes
brspeech	rmvpe_quant	0.0010	2.44e-16	2.44e-16	0.0017	0.0017	yes	yes
brspeech	speaker	0.0010	0	0	0.0207	0.0207	yes	yes
brspeech	wavlm	0.0010	0.3021	0.2685	0.3357	0.0672	yes	yes
brspeech	whisper	0.0010	0.0298	0.0049	0.0547	0.0498	yes	yes
bvcc	auditory_erb	0.0010	9.51e-18	9.51e-18	0.0028	0.0028	yes	yes
bvcc	beats	0.0010	0.2003	0.1609	0.2397	0.0788	yes	yes
bvcc	ced	0.0010	0.2222	0.1783	0.2661	0.0878	yes	yes
bvcc	contentvec12	0.0010	0.0614	0.0228	0.1000	0.0772	yes	yes
bvcc	egemaps	0.0010	0	0	0.0075	0.0075	yes	yes
bvcc	rmvpe_cont	0.0010	0	0	0.0033	0.0033	yes	yes
bvcc	rmvpe_quant	0.0010	1.82e-18	1.82e-18	0.0030	0.0030	yes	yes
bvcc	speaker	0.0010	0.0994	0.0591	0.1397	0.0806	yes	yes
bvcc	wavlm	0.0010	0.1672	0.1244	0.2100	0.0856	yes	yes
bvcc	whisper	0.0010	0.2495	0.2137	0.2853	0.0716	yes	yes
singmos	auditory_erb	0.0010	4.68e-18	4.68e-18	0.0176	0.0176	yes	yes
singmos	beats	0.0010	0.1525	0.1139	0.1912	0.0773	yes	yes
singmos	ced	0.0010	0.2314	0.1831	0.2796	0.0965	yes	yes
singmos	contentvec12	0.0010	0.0761	0.0440	0.1083	0.0643	yes	yes
singmos	egemaps	0.0010	0.2129	0.1606	0.2632	0.1026	yes	yes
singmos	rmvpe_cont	0.0010	1.13e-17	1.13e-17	0.0083	0.0083	yes	yes
singmos	rmvpe_quant	0.0010	4.92e-18	4.92e-18	0.0162	0.0162	yes	yes
singmos	speaker	0.0010	0.0078	0	0.0534	0.0534	yes	yes
singmos	wavlm	0.0010	0.2156	0.1716	0.2597	0.0881	yes	yes
singmos	whisper	0.0010	0.1036	0.0733	0.1339	0.0606	yes	yes

Table S3 — Complete Rashomon intervals (2/4)

Coordinatewise minimum and maximum gate weights among mixtures whose OOF MSE is within the stated relative tolerance of the optimum. All solver-success flags are checked by `verify_release.py`; KKT diagnostics are in `results/rashomon_kkt_audit.csv`.

Corpus	Expert	Tol.	Opt.	Min	Max	Width	Min OK	Max OK
tmhintqi	auditory_erb	0.0010	6.75e-19	6.75e-19	0.0107	0.0107	yes	yes
tmhintqi	beats	0.0010	0.2710	0.1899	0.3521	0.1622	yes	yes
tmhintqi	ced	0.0010	0.2837	0.2002	0.3672	0.1670	yes	yes
tmhintqi	contentvec12	0.0010	0.0716	2.20e-15	0.1442	0.1442	yes	yes
tmhintqi	egemaps	0.0010	0	0	0.0449	0.0449	yes	yes
tmhintqi	rmvpe_cont	0.0010	8.99e-18	8.99e-18	0.0118	0.0118	yes	yes
tmhintqi	rmvpe_quant	0.0010	9.74e-18	9.74e-18	0.0123	0.0123	yes	yes
tmhintqi	speaker	0.0010	0.0118	2.67e-16	0.0769	0.0769	yes	yes
tmhintqi	wavlm	0.0010	0.1697	0.0844	0.2550	0.1707	yes	yes
tmhintqi	whisper	0.0010	0.1922	0.1244	0.2599	0.1355	yes	yes
brspeech	auditory_erb	0.0050	0	0	0.0043	0.0043	yes	yes
brspeech	beats	0.0050	0	0	0.0092	0.0092	yes	yes
brspeech	ced	0.0050	1.67e-16	1.67e-16	0.0552	0.0552	yes	yes
brspeech	contentvec12	0.0050	0.1532	0.0846	0.2218	0.1372	yes	yes
brspeech	egemaps	0.0050	0.5149	0.4235	0.5940	0.1705	yes	yes
brspeech	rmvpe_cont	0.0050	1.08e-16	1.08e-16	0.0317	0.0317	yes	yes
brspeech	rmvpe_quant	0.0050	2.44e-16	2.44e-16	0.0081	0.0081	yes	yes
brspeech	speaker	0.0050	0	0	0.0669	0.0669	yes	yes
brspeech	wavlm	0.0050	0.3021	0.2269	0.3772	0.1503	yes	yes
brspeech	whisper	0.0050	0.0298	9.58e-16	0.0854	0.0854	yes	yes
bvcc	auditory_erb	0.0050	9.51e-18	9.51e-18	0.0132	0.0132	yes	yes
bvcc	beats	0.0050	0.2003	0.1122	0.2884	0.1762	yes	yes
bvcc	ced	0.0050	0.2222	0.1240	0.3204	0.1964	yes	yes
bvcc	contentvec12	0.0050	0.0614	3.37e-15	0.1477	0.1477	yes	yes
bvcc	egemaps	0.0050	0	0	0.0320	0.0320	yes	yes
bvcc	rmvpe_cont	0.0050	0	0	0.0157	0.0157	yes	yes
bvcc	rmvpe_quant	0.0050	1.82e-18	1.82e-18	0.0142	0.0142	yes	yes
bvcc	speaker	0.0050	0.0994	0.0093	0.1896	0.1803	yes	yes
bvcc	wavlm	0.0050	0.1672	0.0715	0.2629	0.1914	yes	yes
bvcc	whisper	0.0050	0.2495	0.1694	0.3296	0.1601	yes	yes

Table S3 — Complete Rashomon intervals (3/4)

Coordinatewise minimum and maximum gate weights among mixtures whose OOF MSE is within the stated relative tolerance of the optimum. All solver-success flags are checked by `verify_release.py`; KKT diagnostics are in `results/rashomon_kkt_audit.csv`.

Corpus	Expert	Tol.	Opt.	Min	Max	Width	Min OK	Max OK
singmos	auditory_erb	0.0050	4.68e-18	4.68e-18	0.0674	0.0674	yes	yes
singmos	beats	0.0050	0.1525	0.0661	0.2390	0.1729	yes	yes
singmos	ced	0.0050	0.2314	0.1235	0.3387	0.2152	yes	yes
singmos	contentvec12	0.0050	0.0761	0.0042	0.1480	0.1439	yes	yes
singmos	egemaps	0.0050	0.2129	0.0960	0.3218	0.2258	yes	yes
singmos	rmvpe_cont	0.0050	1.13e-17	1.13e-17	0.0371	0.0371	yes	yes
singmos	rmvpe_quant	0.0050	4.92e-18	4.92e-18	0.0506	0.0506	yes	yes
singmos	speaker	0.0050	0.0078	0	0.1098	0.1098	yes	yes
singmos	wavlm	0.0050	0.2156	0.1171	0.3141	0.1970	yes	yes
singmos	whisper	0.0050	0.1036	0.0358	0.1713	0.1355	yes	yes
tmhintqi	auditory_erb	0.0050	6.75e-19	6.75e-19	0.0459	0.0459	yes	yes
tmhintqi	beats	0.0050	0.2710	0.0897	0.4515	0.3618	yes	yes
tmhintqi	ced	0.0050	0.2837	0.0970	0.4701	0.3731	yes	yes
tmhintqi	contentvec12	0.0050	0.0716	0	0.2320	0.2320	yes	yes
tmhintqi	egemaps	0.0050	0	0	0.1253	0.1253	yes	yes
tmhintqi	rmvpe_cont	0.0050	8.99e-18	8.99e-18	0.0494	0.0494	yes	yes
tmhintqi	rmvpe_quant	0.0050	9.74e-18	9.74e-18	0.0436	0.0436	yes	yes
tmhintqi	speaker	0.0050	0.0118	2.67e-16	0.1573	0.1573	yes	yes
tmhintqi	wavlm	0.0050	0.1697	1.12e-15	0.3603	0.3603	yes	yes
tmhintqi	whisper	0.0050	0.1922	0.0407	0.3432	0.3025	yes	yes
brspeech	auditory_erb	0.0100	0	0	0.0085	0.0085	yes	yes
brspeech	beats	0.0100	0	0	0.0177	0.0177	yes	yes
brspeech	ced	0.0100	1.67e-16	1.67e-16	0.0882	0.0882	yes	yes
brspeech	contentvec12	0.0100	0.1532	0.0562	0.2503	0.1941	yes	yes
brspeech	egemaps	0.0100	0.5149	0.3771	0.6266	0.2495	yes	yes
brspeech	rmvpe_cont	0.0100	1.08e-16	1.08e-16	0.0585	0.0585	yes	yes
brspeech	rmvpe_quant	0.0100	2.44e-16	2.44e-16	0.0157	0.0157	yes	yes
brspeech	speaker	0.0100	0	0	0.1041	0.1041	yes	yes
brspeech	wavlm	0.0100	0.3021	0.1958	0.4084	0.2126	yes	yes
brspeech	whisper	0.0100	0.0298	9.37e-16	0.1085	0.1085	yes	yes

Table S3 — Complete Rashomon intervals (4/4)

Coordinatewise minimum and maximum gate weights among mixtures whose OOF MSE is within the stated relative tolerance of the optimum. All solver-success flags are checked by `verify_release.py`; KKT diagnostics are in `results/rashomon_kkt_audit.csv`.

Corpus	Expert	Tol.	Opt.	Min	Max	Width	Min OK	Max OK
bvcc	auditory_erb	0.0100	9.51e-18	9.51e-18	0.0254	0.0254	yes	yes
bvcc	beats	0.0100	0.2003	0.0757	0.3249	0.2492	yes	yes
bvcc	ced	0.0100	0.2222	0.0833	0.3611	0.2777	yes	yes
bvcc	contentvec12	0.0100	0.0614	0	0.1834	0.1834	yes	yes
bvcc	egemaps	0.0100	0	0	0.0560	0.0560	yes	yes
bvcc	rmvpe_cont	0.0100	0	0	0.0293	0.0293	yes	yes
bvcc	rmvpe_quant	0.0100	1.82e-18	1.82e-18	0.0268	0.0268	yes	yes
bvcc	speaker	0.0100	0.0994	0	0.2269	0.2269	yes	yes
bvcc	wavlm	0.0100	0.1672	0.0319	0.3026	0.2707	yes	yes
bvcc	whisper	0.0100	0.2495	0.1362	0.3627	0.2265	yes	yes
singmos	auditory_erb	0.0100	4.68e-18	4.68e-18	0.1108	0.1108	yes	yes
singmos	beats	0.0100	0.1525	0.0302	0.2747	0.2444	yes	yes
singmos	ced	0.0100	0.2314	0.0788	0.3828	0.3040	yes	yes
singmos	contentvec12	0.0100	0.0761	4.16e-17	0.1777	0.1777	yes	yes
singmos	egemaps	0.0100	0.2129	0.0458	0.3655	0.3198	yes	yes
singmos	rmvpe_cont	0.0100	1.13e-17	1.13e-17	0.0663	0.0663	yes	yes
singmos	rmvpe_quant	0.0100	4.92e-18	4.92e-18	0.0777	0.0777	yes	yes
singmos	speaker	0.0100	0.0078	0	0.1521	0.1521	yes	yes
singmos	wavlm	0.0100	0.2156	0.0763	0.3548	0.2785	yes	yes
singmos	whisper	0.0100	0.1036	0.0078	0.1994	0.1916	yes	yes
tmhintqi	auditory_erb	0.0100	6.75e-19	6.75e-19	0.0795	0.0795	yes	yes
tmhintqi	beats	0.0100	0.2710	0.0146	0.5256	0.5110	yes	yes
tmhintqi	ced	0.0100	0.2837	0.0196	0.5468	0.5272	yes	yes
tmhintqi	contentvec12	0.0100	0.0716	0	0.2976	0.2976	yes	yes
tmhintqi	egemaps	0.0100	0	0	0.1866	0.1866	yes	yes
tmhintqi	rmvpe_cont	0.0100	8.99e-18	8.99e-18	0.0844	0.0844	yes	yes
tmhintqi	rmvpe_quant	0.0100	9.74e-18	9.74e-18	0.0696	0.0696	yes	yes
tmhintqi	speaker	0.0100	0.0118	2.67e-16	0.2176	0.2176	yes	yes
tmhintqi	wavlm	0.0100	0.1697	9.27e-15	0.4387	0.4387	yes	yes
tmhintqi	whisper	0.0100	0.1922	0	0.4053	0.4053	yes	yes

Table S4 — Nested selection by corpus and budget (1/2)

Equal-system MSE in outer system-disjoint folds. Every rule uses the same cardinality k. Exhaustive is inner-validation search, never oracle selection on the outer test systems. The full 160-row metric table also reports utterance MSE and system SRCC.

Corpus	k	Gate	Singleton	Shapley	Exhaustive
brspeech	1	1.5432	1.5432	1.6283	1.5432
brspeech	2	1.1520	1.1520	1.1520	1.1667
brspeech	3	1.1427	1.1578	1.1543	1.1427
brspeech	4	1.2561	1.1578	1.1506	1.2561
brspeech	5	1.2174	1.1554	1.1112	1.2174
brspeech	6	1.2174	1.2234	1.1829	1.2174
brspeech	7	1.2174	1.1773	1.2336	1.2174
brspeech	8	1.2174	1.2336	1.2336	1.2174
brspeech	9	1.2174	1.2174	1.2174	1.2174
brspeech	10	1.2174	1.2174	1.2174	1.2174
bvcc	1	0.3437	0.3338	0.3338	0.3338
bvcc	2	0.2820	0.2864	0.2918	0.2778
bvcc	3	0.2595	0.2654	0.2621	0.2585
bvcc	4	0.2497	0.2549	0.2514	0.2497
bvcc	5	0.2477	0.2474	0.2477	0.2477
bvcc	6	0.2469	0.2469	0.2469	0.2469
bvcc	7	0.2469	0.2469	0.2469	0.2469
bvcc	8	0.2469	0.2469	0.2469	0.2469
bvcc	9	0.2469	0.2469	0.2469	0.2469
bvcc	10	0.2469	0.2469	0.2469	0.2469

Table S4 — Nested selection by corpus and budget (2/2)

Equal-system MSE in outer system-disjoint folds. Every rule uses the same cardinality k. Exhaustive is inner-validation search, never oracle selection on the outer test systems. The full 160-row metric table also reports utterance MSE and system SRCC.

Corpus	k	Gate	Singleton	Shapley	Exhaustive
singmos	1	0.4779	0.4779	0.4692	0.4779
singmos	2	0.4179	0.4105	0.4041	0.4041
singmos	3	0.3877	0.3842	0.3869	0.3921
singmos	4	0.3801	0.3844	0.3801	0.3793
singmos	5	0.3745	0.3819	0.3813	0.3739
singmos	6	0.3762	0.3814	0.3797	0.3749
singmos	7	0.3742	0.3797	0.3781	0.3742
singmos	8	0.3744	0.3796	0.3787	0.3744
singmos	9	0.3744	0.3770	0.3768	0.3744
singmos	10	0.3744	0.3744	0.3744	0.3744
tmhintqi	1	0.9986	0.9754	0.9754	0.9754
tmhintqi	2	0.9341	0.9272	0.9356	0.9363
tmhintqi	3	0.9158	0.9171	0.9138	0.9127
tmhintqi	4	0.9034	0.9042	0.9024	0.9034
tmhintqi	5	0.8993	0.9010	0.8993	0.9017
tmhintqi	6	0.8999	0.9011	0.8999	0.8999
tmhintqi	7	0.9001	0.9001	0.9001	0.9001
tmhintqi	8	0.9001	0.9001	0.9001	0.9001
tmhintqi	9	0.9001	0.9001	0.9001	0.9001
tmhintqi	10	0.9001	0.9001	0.9001	0.9001

Table S5 — Null: permuted redundancy (1/1)

Complete numeric outcomes for this null family. Protocol notes remain verbatim in results/null_experiments.csv. These controls diagnose how much of the correlation–interaction pattern follows from regression geometry rather than speech-specific structure.

experiment	dataset	true rho corr vs neg	null mean	null sd	null p025	null p975	emp p null ge true	n null
permuted_redundan	brspeech	0.3681	-0.3806	0.0413	-0.4621	-0.2957	0.0010	1.00e+03
permuted_redundan	bvcc	0.9431	-0.8460	0.0137	-0.8722	-0.8199	0.0010	1.00e+03
permuted_redundan	singmos	0.4432	-0.3411	0.0357	-0.4053	-0.2661	0.0010	1.00e+03
permuted_redundan	tmhintqi	0.9574	-0.8453	0.0098	-0.8664	-0.8288	0.0010	1.00e+03

Table S5 — Null: covariance matched gaussian (1/1)

Complete numeric outcomes for this null family. Protocol notes remain verbatim in results/null_experiments.csv. These controls diagnose how much of the correlation–interaction pattern follows from regression geometry rather than speech-specific structure.

experiment	dataset	n pairs	feasible	rho obs corr vs negl	rho matched null
covariance_matched_gaussian	brspeech	45.0000	15.0000	0.3097	0.3016
covariance_matched_gaussian	bvcc	45.0000	41.0000	0.9578	0.9565
covariance_matched_gaussian	singmos	45.0000	42.0000	0.4279	0.8013
covariance_matched_gaussian	tmhintqi	45.0000	26.0000	0.9723	0.8852

Table S5 — Null: residual controls (1/1)

Complete numeric outcomes for this null family. Protocol notes remain verbatim in results/null_experiments.csv. These controls diagnose how much of the correlation–interaction pattern follows from regression geometry rather than speech-specific structure.

experiment	dataset	beta corr	beta mean	beta min
residual_controls	brspeech	0.2342	0.9432	-0.1616
residual_controls	bvcc	0.3771	0.6984	-0.1368
residual_controls	singmos	0.0475	1.0525	-0.1467
residual_controls	tmhintqi	0.5501	0.5294	-0.1234

Table S5 — Null: full pipeline (1/1)

Complete numeric outcomes for this null family. Protocol notes remain verbatim in results/null_experiments.csv. These controls diagnose how much of the correlation–interaction pattern follows from regression geometry rather than speech-specific structure.

experiment	dataset	variant	observed nzero	observed count	null nzero mean	null nzero max	null count mean	null count max	null conv rate	n null
full_pipeline	brspeech	eigclip	6.0000	5.0000	0	0	0	0	1.0000	5.0000
full_pipeline	bvcc	eigclip	4.0000	4.0000	0	0	0	0	1.0000	5.0000
full_pipeline	singmos	eigclip	3.0000	3.0000	0	0	0	0	1.0000	5.0000
full_pipeline	tmhintqi	eigclip	4.0000	4.0000	0	0	0	0	1.0000	5.0000
full_pipeline	brspeech	higham	6.0000	5.0000	0	0	0	0	1.0000	5.0000
full_pipeline	bvcc	higham	4.0000	4.0000	0	0	0	0	1.0000	5.0000
full_pipeline	singmos	higham	3.0000	3.0000	0	0	0	0	1.0000	5.0000
full_pipeline	tmhintqi	higham	4.0000	4.0000	0	0	0	0	1.0000	5.0000

Table S6 — Interaction-correlation uncertainty (1/1)

Correlation of expert-prediction similarity with negative Shapley interaction. The test correlation interval resamples test systems while experts and fitted gates remain fixed; this is descriptive rather than full-refit uncertainty.

Corpus	Systems	OOF rho	OOF low	OOF high	Test rho	Test low	Test high	B
brspeech	6	0.3243	0.0167	0.4801	0.5995	0.1161	0.7794	2000
bvcc	187	0.9040	0.8386	0.9310	0.9634	0.9107	0.9789	2000
singmos	35	0.3261	0.1233	0.7524	0.3354	0.1853	0.7444	2000
tmhintqi	97	0.9377	0.8087	0.9508	0.9568	0.8607	0.9743	2000

Artifact map and interpretation boundaries

Allocation: results/all_40_cells.csv and results/discordant_cases.csv contain point estimates, system intervals, and refit frequencies. Contextual utility: results/shapley_values.csv and results/interaction_values.csv contain exact coalition-game quantities. Near-optimal allocation: results/rashomon_intervals.csv and results/rashomon_kkt_audit.csv contain every coordinate extreme and its solver diagnostics.

Selection: results/nested_selections.csv contains every outer-fold selected subset and fitted weight vector; results/nested_selection_metrics.csv contains every corpus × rule × budget outcome; results/nested_selection_bootstrap.csv contains macro contrasts and Holm corrections. Exhaustive means best inner-validation subset, not best outer-test subset.

Nulls: results/null_experiments.csv records four control families, and results/interaction_bootstrap.csv records uncertainty for the observed redundancy relation. The matched Gaussian control reproduces much of the association, so the artifact supports a geometric explanation and does not claim a uniquely speech-specific law.

Scope: Shapley quantifies contextual utility under the declared player partition; it is not a universal selection rule. Singleton ranking was the attractive low-cost operating point in this benchmark, but no equivalence or universal encoder ranking is claimed. BRSpeechMOS system-level inference is exploratory because it has six systems.

Release identity: the version is embedded above and metadata/artifact_manifest.json binds tracked files to SHA-256 digests. The Git tag binds that manifest to a commit. The DOI field must be updated only after Zenodo archives the final GitHub Release.